

DATA ANALYSIS INTERNSHIP

Day 3 – REPORT

Data Cleaning and Preprocessing for Analysis (Home Assignment)

On Day 3, **Data Cleaning and Preprocessing techniques** were performed using the **Student Performance Dataset** that was analyzed during **Exploratory Data Analysis (EDA)**. The preprocessing process was carried out to improve dataset quality and prepare the data for further analysis and model building.

1. Dataset Verification

The **Student Performance Dataset** was verified carefully to ensure that all student records were properly stored and organized.

The following details were checked:

- Student ID values
- Hours studied
- Sleep hours
- Attendance percentage
- Previous scores
- Exam score
- Result categories (**Pass/Fail**)

Observation:

- The dataset contained both **student behavioral details** and **academic performance records** required for analysis.
 - Proper organization of records improved dataset readability and understanding.
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2. Missing Value Checking

The dataset was checked for **missing or empty values** in important columns to maintain data quality.

Columns Verified:

- hours_studied
- sleep_hours
- attendance_percent
- previous_scores
- exam_score
- result

Observation:

- Missing values can affect analysis accuracy and machine learning performance.
 - Proper verification helps maintain dataset consistency and reliability.
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3. Duplicate Value Checking

Duplicate records were verified to avoid repeated student entries inside the dataset.

Verified Areas:

- student_id
- Student performance records
- Result values

Observation:

- Duplicate records may produce incorrect analysis results.
 - Removing duplicate entries improves dataset reliability and accuracy.
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4. Column Organization

Unwanted columns were removed and only the required columns were retained for analysis.

Final Columns Used:

- student_id
- hours_studied
- sleep_hours
- attendance_percent
- previous_scores
- exam_score
- result

Observation:

Organizing necessary columns makes the dataset easier to analyze and improves readability.

5. Data Type Verification

The data types of each column were verified to ensure proper format and consistency.

Column	Data Type
student_id	String
hours_studied	Float
sleep_hours	Float
attendance_percent	Float
previous_scores	Integer
exam_score	Float
result	String

Observation:

Correct data types help avoid calculation errors and improve preprocessing quality.

6. Data Validation

The dataset values were validated to ensure meaningful and properly formatted data.

Validated Details:

- Study hour values
- Sleep hour values
- Attendance percentage values
- Previous academic scores
- Exam score values
- Result categories (**Pass/Fail**)

Observation:

- Data validation helps avoid incorrect analysis and improves consistency.

- Proper validation ensures meaningful results during prediction and classification.
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7. Data Preprocessing for Analysis

Preprocessing techniques were applied to prepare the dataset for future machine learning analysis.

Steps Performed:

- Verified missing values
- Checked duplicate records
- Organized required columns
- Validated numerical values
- Verified categorical values (**Pass/Fail**)
- Prepared clean data for analysis

Observation:

A clean dataset improves prediction accuracy and model performance.

8. Conclusion

On Day 3, I learned the importance of **Data Cleaning and Data Preprocessing** in **Data Analysis**. I verified missing values, checked duplicate records, organized important columns, validated data types, and cleaned the **Student Performance Dataset** for further analysis. These preprocessing techniques improved the **quality, consistency, and reliability** of the dataset and prepared it for future analysis and machine learning tasks.